When LLMs Revise Beyond the User-Defined Threshold: Prompt Phrasing Controls the Revision Gate

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Abstract

Official prompting guidance emphasizes defining explicit success criteria and output constraints. Users follow this advice, routinely embedding quality thresholds in prompts ("this only needs to be a 70 out of 100") and assuming this constrains model revision behavior. We test this assumption in a two-turn writing-revision setup with 3,840 trials crossing follow-up prompt phrasing (2 levels), stated quality threshold (8 levels), and model (GPT-4o, Claude Sonnet 4, Gemini 2.5 Flash) across eight scenarios. We find that quality thresholds have *no detectable effect* on whether the model revises at all. A revision-implying follow-up ("Can this be improved?") triggers near-universal revision (99.9%) regardless of threshold, while an evaluative follow-up produces 0.3–38.0% revision rates depending on model. Within triggered revisions, thresholds produce only a weak-to-moderate dose-response on overcorrection (Spearman $\rho = -0.14$ to -0.50). These results are consistent with a **two-stage account**: Stage 1 (revision gate) is controlled by prompt phrasing; Stage 2 (revision intensity) is modulated by the stated threshold. The two stages respond to different input signals, and a significant probe $\times$ threshold interaction is consistent with a floor effect rather than true coupling. The practical consequence is that models possess threshold-calibration ability but gate it behind a phrasing-sensitive decision with no detectable threshold sensitivity, suggesting that user-stated thresholds may be largely ineffective under revision-implying conversational flows.

1 Introduction

Official prompting guidance from major model providers encourages users to define explicit success criteria and output constraints. OpenAI's prompting documentation recommends defining "the output contract" with exact deliverables and expected outcomes (OpenAI, 2025); Anthropic's best-practices guide emphasizes being "clear and direct" about desired output and specifying constraints up front (Anthropic, 2025). The implication is intuitive: if you tell a model that a task only needs to be a 70 out of 100, it should produce a quick, functional draft rather than an exhaustive opus. Users internalize this assumption and routinely embed quality thresholds in their prompts, treating them as binding constraints on model behavior (White et al., 2023; Schulhoff et al., 2024).

But does the model actually respect these constraints when asked to revise? Consider an illustrative case: a user asks for a quick text message canceling brunch plans, specifying the quality only needs to be "around a 70 out of 100." The model produces a reasonable two-sentence text. The user then asks, "Can this be improved?" Instead of a minor polish, the model returns a three-paragraph message with a greeting, an apology, and a rain-check proposal. The stated threshold was 70; the revision overshoots it dramatically. This pattern, where models revise far beyond the user's stated quality bar, is what we call *overcorrection*, and we observe it across all eight scenarios in our experiment.

We test this empirically in a two-turn writing-revision setup. In a fully factorial experiment with 3,840 trials, we cross follow-up prompt phrasing (2 levels), stated quality threshold (8 levels, with numeric and qualitative framings), and model (GPT-4o, Claude Sonnet 4, Gemini 2.5 Flash) across eight writing scenarios spanning four register categories. An LLM judge evaluates each revision on a revision gate classification and four rubric dimension scores, with inter-rater reliability verified against a second judge model.

We find that the follow-up prompt's phrasing, not the quality threshold, is the dominant predictor of whether the model revises. A five-probe pilot study reveals this starkly: revision rates drop

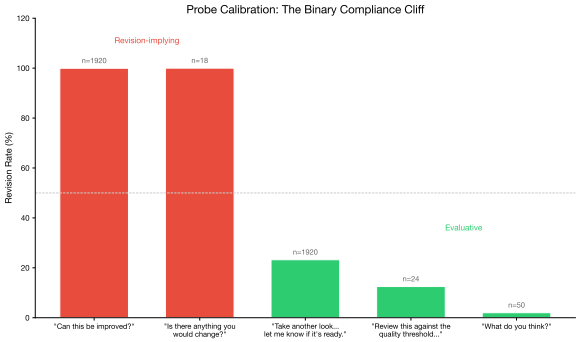

Figure 1: The compliance cliff across five pilot probes. Revision rate drops from 100% to $\leq$25% with no smooth gradient, indicating a categorical rather than continuous sensitivity to prompt phrasing. This binary pattern motivates the two-stage account developed in the paper.

from 100% to below 25% based on probe phrasing alone, with no smooth gradient between (Figure 1). Thresholds do modulate *how much* the model overcorrects once revision is triggered, but this calibration ability is locked behind a phrasing-sensitive gate that shows no detectable sensitivity to thresholds (Kruskal-Wallis $p > 0.40$ for all models). We characterize this as a **two-stage account**: Stage 1 is a phrasing-sensitive *revision gate*; Stage 2 is a threshold-sensitive *intensity calibration*. The calibration ability users would most benefit from (having the model decline unnecessary revision) is not activated under the revision-implying conversational conditions we tested.

Our contributions are:

1. A two-stage account (gate + intensity) of LLM revision behavior, supported by 3,840 trials showing that the decision to revise and the calibration of that revision respond to different input signals, with a significant but floor-effect-driven interaction.

2. A five-probe calibration study documenting a steep *compliance cliff*: revision rates shift from near-100% to near-0% based on probe phrasing, with no observed gradient between (though 5 probes cannot rule out a steep continuous function).

3. A cross-model comparison showing that revision disposition under evaluative prompts varies sharply across model families (0.3% to 38.0% revision rate), while behavior under revision-implying prompts is uniformly compliant.

2 Related Work

Sycophancy and Instruction Compliance. Instruction-tuned LLMs exhibit a well-documented tendency toward sycophantic behavior: agreeing with users, complying with requests, and avoiding disagreement even when it would be more helpful (Perez et al., 2023; Sharma et al., 2024; Wei et al., 2024). Ranaldi and Pucci (2023) show that this tendency extends to settings where models comply with implied requests even when the user's own stated preferences suggest otherwise. Our work documents a specific pattern consistent with this compliance tendency in revision tasks: a phrasing-sensitive revision gate that fires regardless of the user's stated quality standard.

RLHF and Alignment Training. The revision gate's bias toward compliance is plausibly shaped by reinforcement learning from human feedback (RLHF), in which reward models trained on human preference data may systematically reward helpful action over restraint (Ouyang et al., 2022; Bai et al., 2022). Direct preference optimization (Rafailov et al., 2023) inherits this bias to the extent that training pairs favor responses that "do something" over responses that decline. Our findings are consistent with this hypothesis: all three models, despite different training pipelines, exhibit near-universal compliance under revision-implying prompts.

Prompt Sensitivity. LLMs are highly sensitive to surface-level features of prompts (phrasing, formatting, and ordering), often more than to semantic content (Lu et al., 2022; Zhao et al., 2021; Sclar et al., 2024). Mizrahi et al. (2024) demonstrate that minor reformulations of equivalent instructions produce substantially different outputs. We extend this literature to the revision setting, showing that the pragmatic force of a follow-up prompt (revision-implying vs. evaluative) dominates a semantically meaningful constraint (the quality threshold). Notably, this is a multi-turn phenomenon: the threshold is stated in Turn 1 but the gate-controlling signal arrives in Turn 2, raising the possibility that recency weighting contributes to threshold insensitivity.

LLM Self-Refinement. A growing body of work explores LLMs' ability to iteratively refine their own outputs. Madaan et al. (2024) propose Self-Refine, showing that models can improve

outputs through prompted self-critique. Huang et al. (2024) argue that models cannot reliably self-correct reasoning without external feedback. Kim et al. (2024) examine conditions under which self-refinement succeeds or fails in reasoning tasks. These studies typically assume that refinement is desirable; we examine the complementary question of whether models can *refrain* from refinement when the user's quality standard has already been met.

LLM-as-Judge. Using LLMs to evaluate generated text has become standard practice (Zheng et al., 2024; Li et al., 2024). Concerns about self-preferencing bias, where a model favors its own outputs, are well-documented (Panickssery et al., 2024; Koo et al., 2024). We address this by testing for judge bias directly, using a second judge model for inter-rater reliability, and employing a judge-independent response length proxy that is broadly consistent with judge rankings.

Controllable Generation and Quality Calibration. Holistic evaluation efforts such as HELM (Liang et al., 2023) have assessed model capabilities across style, toxicity, and formality dimensions, but the question of whether models calibrate *revision behavior* to user-stated quality thresholds has received limited direct attention. Zhou et al. (2024) study calibration in the context of next-token prediction, and Lin et al. (2022) explore models' ability to express uncertainty. Our work is, to our knowledge, the first to empirically test whether the decision to revise (not just the quality of the output) is sensitive to user-stated standards.

3 Methods

3.1 Task Design

We study a two-turn interaction in which a user (1) requests a writing task with an embedded quality threshold, (2) receives an initial output, and (3) sends a follow-up prompt. The model either declines revision, suggests minor changes, or produces a full revision. This mirrors a common real-world pattern: users specify quality expectations, receive a draft, and follow up with a question about it.

3.2 Independent Variables

The experiment crosses three factors in a fully factorial design.

Probe Framing (2 levels). The follow-up prompt after the model's initial output:

- **Leading:** "Can this be improved?" Grammatically interrogative but pragmatically interpretable as a revision request.
- **Evaluative:** "Take another look at this and let me know if it's ready." Invites assessment without directing revision.

Quality Threshold (8 levels $\times$ 2 framings). The user's stated quality bar, embedded in the initial prompt. Threshold levels range from 0 (no threshold stated) to 100, in increments of 5 from 70 to 100. Each level has two framing variants:

- **Numeric:** "I would consider this done if the outcome is around a [X] out of 100."
- **Qualitative:** Functionally equivalent natural language (e.g., 70 $\rightarrow$ "good enough to get the point across without causing any problems").

Model (3 levels). GPT-4o (OpenAI), Claude Sonnet 4 (Anthropic), Gemini 2.5 Flash (Google).

3.3 Probe Selection

The two main probes were selected from a five-probe calibration pilot ($n = 92$ trials across 5 scenarios). The pilot revealed a *binary compliance cliff* (Figure 1): probes partition into two categories with no intermediate region.

Any probe implying revision (Leading: "Can this be improved?", 100%; Pilot A: "Is there anything you would change?", 100%) triggered near-universal compliance. Any probe implying evaluation (Neutral: "What do you think?", 2%; Pilot B: "Review this against the threshold", 12.5%) triggered near-zero revision. Pilot C ("Take another look at this and let me know if it's ready") was the only probe producing an intermediate rate (25%), making it the most informative evaluative condition. We selected Leading and Pilot C as the two main probes.

3.4 Scenarios

Eight writing scenarios span four register categories (Table 1). Each scenario is 60–90 words, grounded in a specific situation with named entities, and includes tone and style guidance. The full design crosses 8 scenarios $\times$ 2 framings $\times$ 8 thresholds $\times$ 2 probes $\times$ 3 models $\times$ 5 runs $=$ **3,840 primary trials**, plus 92 pilot trials (3,932 total scored).

Register	ID	Description
Formal	pto	PTO request to manager
Formal	sales	Sales email to outside client
Neutral	linkedin	LinkedIn job announcement
Neutral	slack	Slack project status update
Informal	brunch	Text canceling brunch plans
Informal	coworker	Funny text to a coworker
Technical	setup	Software setup instructions
Creative	review	Online headphones review

Table 1: Eight writing scenarios across four register categories.

Dimension	QW κ	Interpretation
Revision magnitude	0.844	Excellent
Revision value	0.690	Good
Overcorrection	0.609	Acceptable
Threshold alignment	0.556	Marginal

Table 2: Inter-rater reliability (quadratic-weighted Cohen's κ) between GPT-4o and Claude Sonnet 4 judges on a stratified sample of 60 trials.

3.5 Evaluation

All trials are scored by GPT-4o as an LLM judge using a structured rubric. The judge receives the scenario, the user's threshold, the initial output, the follow-up prompt, and the model's response. It returns a JSON object with five scored outputs: a revision gate classification (3-level categorical) and four continuous dimension scores.

Dependent Variables. The *revision gate* is a three-level categorical variable: decline, suggest minor changes, or full revision. For RQ2, we measure *overcorrection* on a 1–5 scale, where 1 indicates proportionate revision and 5 indicates severe overcorrection relative to the stated threshold. Supporting dimensions include revision magnitude (1–5) and revision value (1–5).

Inter-Rater Reliability. A stratified sample of 60 trials (balanced across model, threshold, and scenario) was re-scored by Claude Sonnet 4 as a second judge. Table 2 reports quadratic-weighted Cohen's κ. Agreement is strongest on revision magnitude and weakest on threshold alignment, which readers should weigh when interpreting threshold-related findings.

Bias Controls. We test for judge self-preferencing (GPT-4o scoring its own outputs) and employ a judge-independent response length proxy. Between-model rankings on the length proxy are broadly consistent with judge rankings at the between-model level, though the proxy is not uniformly reliable within model (§4.5).

3.6 Analysis

Figure 2 provides an overview of the full experimental pipeline.

We use chi-squared tests for revision gate distributions (RQ1), Spearman correlations and Kruskal-Wallis H tests for threshold–overcorrection relationships (RQ2), and ordinal regression for multivariate analysis. We fit three ordinal regression models:

- **Model A:** overcorrection $\sim$ probe $+$ threshold $+$ framing $+$ model
- **Model B:** Model A $+$ scenario fixed effects
- **Model C:** Leading-probe trials only: overcorrection $\sim$ threshold $+$ framing $+$ model $+$ scenario

All p-values are corrected for multiple comparisons using Benjamini-Hochberg FDR ($q < 0.05$).

4 Results

Our results are consistent with a two-stage account of revision behavior, visualized in Figure 3. Stage 1 (the revision gate) responds to probe phrasing; Stage 2 (intensity calibration) responds to the stated threshold. We present evidence for each stage in turn.

4.1 RQ1: The Revision Gate

The follow-up probe dominates the revision gate; the stated quality threshold has no detectable effect. Under the leading probe ("Can this be improved?"), models revised in 99.9% of trials: GPT-4o and Claude Sonnet 4 declined 0.0% of the time, and Gemini 2.5 Flash declined just 0.2%. Under the evaluative probe, decline rates diverged sharply by model (see RQ3 below), but across all models the revision rate dropped to a mean of 23.2%.

Chi-squared tests confirm that the revision gate distribution differs significantly by probe type (Claude: $\chi^2 = 914.37$, $p < 0.0001$; Gemini: $\chi^2 = 1326.89$, $p < 0.0001$; GPT-4o: $\chi^2 =$

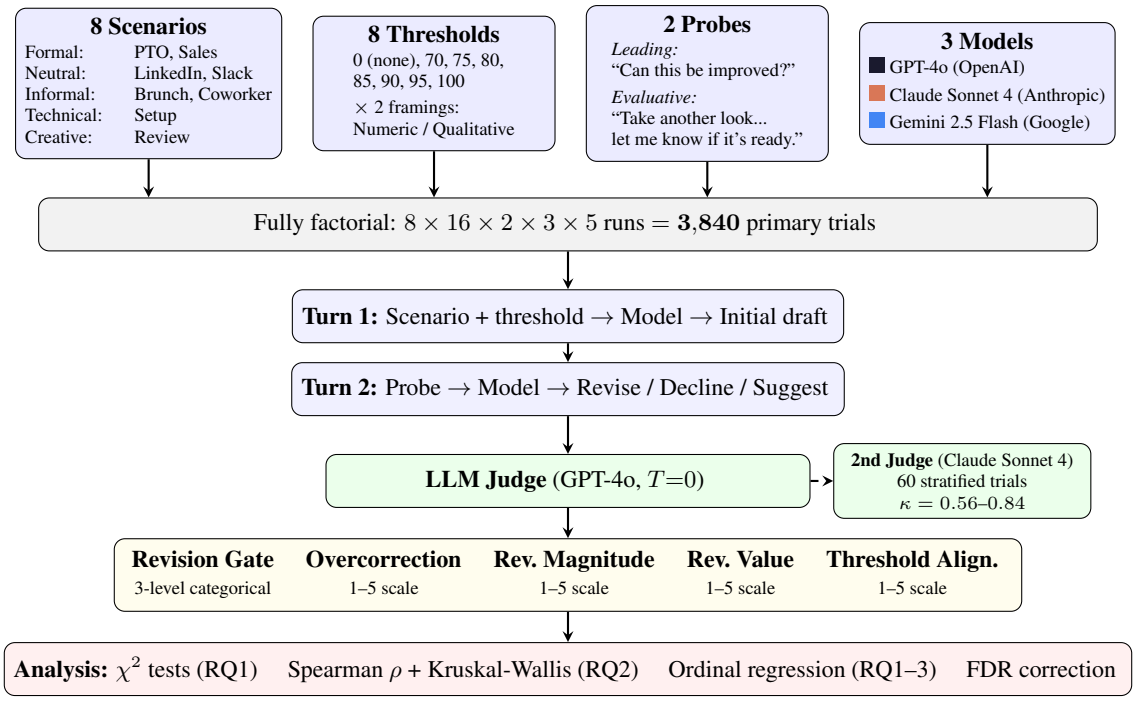

Figure 2: Experimental pipeline overview. Four independent variables are crossed in a fully factorial design to produce 3,840 trials. Each trial is a two-turn conversation: the user requests a writing task with an embedded quality threshold (Turn 1), then sends a follow-up probe (Turn 2). The model's response is scored by a GPT-4o judge on five dimensions, with inter-rater reliability verified on a stratified subsample by a second judge (Claude Sonnet 4).

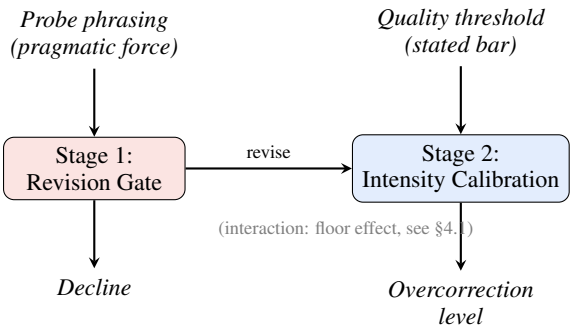

Figure 3: Two-stage account of revision behavior. Stage 1 (revision gate) is controlled by the pragmatic force of the follow-up prompt; Stage 2 (intensity calibration) is modulated by the user's stated quality threshold. The probe × threshold interaction is significant but consistent with a floor effect (§5). Within each probe condition, the stages appear to operate on separate inputs.

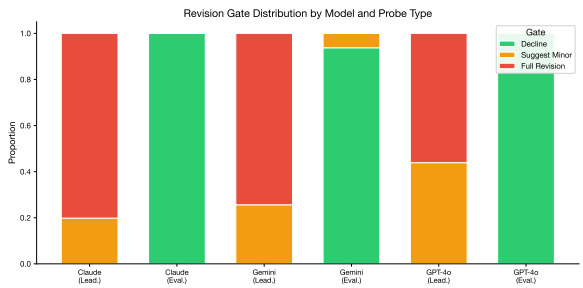

Figure 4: Revision gate distribution by model and probe type. The leading probe produces near-universal revision across all models. The evaluative probe suppresses revision, with model-specific decline rates ranging from 62% (Claude) to 99.7% (Gemini). See Appendix Figure 9 for the corresponding overcorrection comparison.

788.96, $p < 0.0001$).[1] The quality threshold, by contrast, does not significantly affect the revision gate distribution for any model (Kruskal-Wallis $p > 0.40$ for all models). Given the study's power to detect proportion differences as small as 0.032, these null results are unlikely to reflect insufficient power.

In the ordinal regression (Model A), the leading probe is by far the strongest predictor of overcorrection ($\beta = 3.47$, $z = 33.73$, $p < 0.001$), dwarfing all other coefficients (Table 3). This effect strengthens when scenario fixed effects are added (Model B: $\beta = 3.94$, $p < 0.001$). Adding a probe

[1] These chi-squared tests include all five pilot probe types, not just the two main probes; degrees of freedom reflect the full 5-probe × 3-category contingency table.

$\times$ threshold interaction term yields a significant coefficient ($\beta = -2.20$, $z = -5.69$, $p < 0.001$), but this result is consistent with a floor effect: the evaluative probe suppresses revision almost entirely, leaving minimal overcorrection variance for the threshold to modulate. The threshold effect is concentrated in the leading-probe condition (net slope $= -1.40$), while the evaluative-probe slope ($+0.80$) is driven by the small minority of trials that pass the gate.

4.2 RQ2: Threshold Sensitivity Within Revisions

Within the leading-probe condition, where revision is universal, higher stated thresholds are associated with lower overcorrection scores, but the effect size varies substantially across models (Figure 5). Median overcorrection is 1.0 across all conditions, indicating that most individual trials show proportionate revision; the regression effects below are driven by a substantial minority of trials with moderate-to-severe overcorrection, concentrated in formal and creative writing scenarios (see Appendix Figure 13).

Spearman correlations between threshold level and overcorrection (leading probe only) are:

- Gemini 2.5 Flash: $\rho = -0.50$ ($p < 0.001$), moderate
- Claude Sonnet 4: $\rho = -0.35$ ($p < 0.001$), weak-to-moderate
- GPT-4o: $\rho = -0.14$ ($p < 0.001$), negligible-to-weak

Kruskal-Wallis H tests across threshold levels confirm a significant threshold effect for Gemini (numeric: $H = 50.40$, $p < 0.0001$; qualitative: $H = 23.42$, $p = 0.001$) and Claude (numeric: $H = 17.59$, $p = 0.014$; qualitative: $H = 23.30$, $p = 0.002$), but not for GPT-4o (numeric: $H = 3.20$, $p = 0.87$; qualitative: $H = 4.89$, $p = 0.67$).

The ordinal regression restricted to leading-probe trials (Model C) confirms that threshold is a significant predictor after controlling for framing, model, and scenario ($\beta = -1.65$, $p < 0.001$). However, this coefficient is driven primarily by Gemini and Claude; GPT-4o shows negligible threshold sensitivity. For the cell-level overcorrection heatmap, see Appendix Figure 10.

4.3 RQ3: Model Differences in Revision Disposition

Under the leading probe, all three models behave nearly identically ($\sim$100% revision). The evaluative probe reveals sharply different revision dispositions:

- Gemini 2.5 Flash: 99.7% decline rate (almost never revises)
- GPT-4o: 68.6% decline rate
- Claude Sonnet 4: 62.0% decline rate (most willing to revise)

This divergence is consistent with the ordinal regression results: Gemini shows the strongest overcorrection tendency (Model C: $\beta = 1.15$, $p < 0.001$) but also the highest evaluative-probe decline rate. When Gemini does revise, it overcorrects more aggressively, but it is also the model most likely to correctly decline revision under an evaluative prompt. This pattern suggests that Gemini's revision gate is more sharply tuned: it commits fully in either direction, while GPT-4o and Claude show more "leakage" between modes (Figure 6; see also Appendix Figure 16).

4.4 Framing Effects

Qualitative threshold framings produce significantly less overcorrection than numeric framings (Model A: $\beta = -0.74$, $p < 0.001$; Model C: $\beta = -1.05$, $p < 0.001$). This effect is consistent across all three regression models and is not an artifact of scenario or model differences. Revision gate distributions do not differ by framing (χ^2 tests: all $p > 0.77$), confirming that the framing effect operates at Stage 2 (intensity), not Stage

Predictor	Model A	Model B	Model C
is_leading	3.47^{***}	3.94^{***}	—
threshold	-0.93^{***}	-1.04^{***}	-1.65^{***}
is_qualitative	-0.74^{***}	-0.85^{***}	-1.05^{***}
Claude Sonnet	0.10	0.15	0.33^{**}
Gemini Flash	0.42^{***}	0.52^{***}	1.15^{***}
Scenario FEs	No	Yes	Yes
N	3,840	3,840	1,920
AIC	5,319	4,839	3,545

Table 3: Ordinal regression coefficients (overcorrection $\sim$ predictors). Model C is restricted to leading-probe trials. Threshold is scaled 0–1 (so $\beta = -1.65$ corresponds to the full range from no threshold to threshold 100). $^{**}p < 0.01$, $^{***}p < 0.001$. Reference model: GPT-4o.

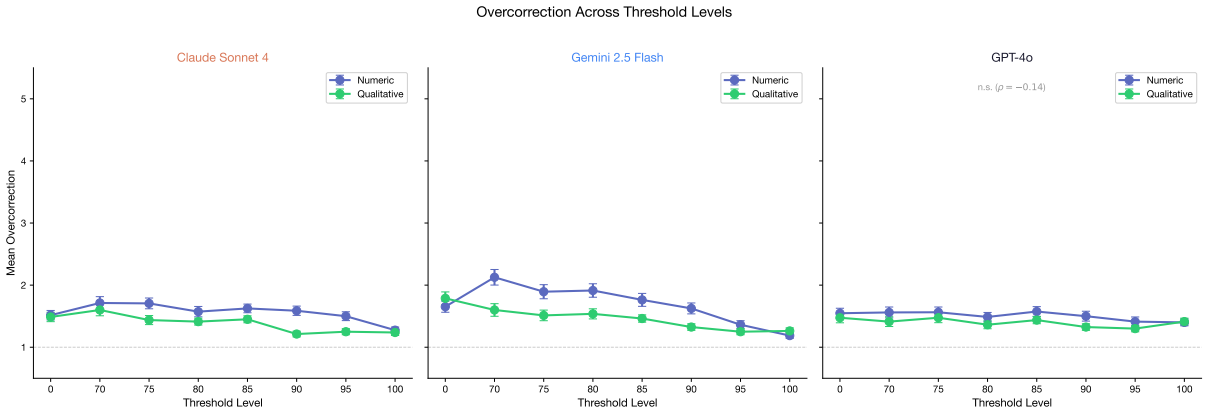

Figure 5: Overcorrection by threshold level within leading-probe trials (one panel per model). Higher thresholds are associated with less overcorrection for Gemini ($\rho = -0.50$) and Claude ($\rho = -0.35$), but the effect is not statistically distinguishable from zero for GPT-4o ($\rho = -0.14$). Dashed line marks proportionate revision (overcorrection $= 1$).

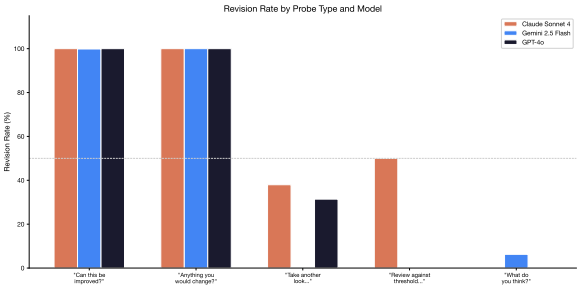

Figure 6: Compliance cliff broken down by model. All three models exhibit the same binary cliff pattern, but the evaluative-probe decline rate varies sharply: Gemini declines 99.7% of the time, while Claude and GPT-4o show more "leakage" into revision mode.

1 (gate). For the full framing comparison, see Appendix Figures 12 and 11.

4.5 Robustness

Length Proxy. Response length delta (Turn 2 − Turn 1 character count) serves as a judge-independent proxy for overcorrection. Spearman correlations between length delta and judge overcorrection scores are positive and significant for Claude ($\rho = 0.20$, $p < 0.001$) and Gemini ($\rho = 0.56$, $p < 0.001$). GPT-4o shows a near-zero negative correlation ($\rho = -0.07$, $p = 0.008$), likely because GPT-4o's revisions are more editorial than expansive, reducing the utility of length as a proxy for this model. Between-model rankings on length delta are broadly consistent with judge rankings on overcorrection (Figure 7): Gemini shows the largest length increases, Claude intermediate, and GPT-4o the smallest. See also Appendix Figure 15 for the full length delta breakdown.

Judge Self-Preferencing. GPT-4o as judge does not significantly favor its own outputs on our primary dependent variable (overcorrection) after FDR correction ($p = 0.032 \rightarrow q = 0.055$). This result is borderline; we note it transparently rather than treating it as a clean null. On revision value, the judge scores GPT-4o outputs *lower* than other models (self mean $= 1.84$ vs. other mean $= 2.27$), suggesting a conservative rather than self-preferencing bias. See Appendix Figure 14 for the full comparison.

Power. Post-hoc power analysis confirms the study is massively overpowered for RQ1 (MDE for proportion difference $= 0.032$; observed ≈ 0.75) and well-powered for RQ2 (MDE for $|\rho| = 0.064$; observed 0.14–0.50).

5 Discussion

The Two-Stage Account. The results presented above are consistent with a two-stage account of LLM revision behavior (Figure 3) in which the decision to revise and the calibration of that revision respond to different input signals. Stage 1 (the revision gate) responds to the pragmatic force of the follow-up prompt with near-binary sensitivity: any phrasing that implies revision triggers compliance; any phrasing that implies evaluation suppresses it. Stage 2 (revision intensity) responds to the user's stated quality threshold with a weak-to-moderate dose-response. The probe × threshold interaction term is statistically significant ($\beta = -2.20$, $z = -5.69$, $p < 0.001$), but this is consistent with a floor effect rather than true coupling: because the evaluative probe suppresses revision al-

most entirely, there is no overcorrection variance for the threshold to modulate. Within each probe condition taken separately, the two stages appear to operate on separate inputs: the threshold has no detectable effect on the gate (Kruskal-Wallis $p > 0.40$), and probe phrasing does not modulate calibration intensity within the leading-probe condition.

This pattern is the central observation of the paper. It means that the calibration ability demonstrated by Stage 2, which shows models *do* modulate revision intensity based on thresholds, is effectively inaccessible at Stage 1 under the conversational conditions we tested. Users who state quality preferences and then ask casual follow-up questions receive revisions that are insensitive to their stated standards, not because the model lacks calibration ability, but because the gate fires before calibration is reached.

Threshold-Blind Compliance. A natural objection is that "Can this be improved?" is a leading question with a pragmatically obvious answer: almost anything *can* be improved. A model that answers "yes" and demonstrates an improvement may be following instructions appropriately, not behaving sycophantically. Whether the gate reflects sycophancy or straightforward instruction-following, the key observation is the same: the gate is threshold-blind. A well-calibrated instruction-follower would incorporate the user's stated quality bar: if the initial output meets a threshold of 70, and the user asks whether it can be improved, the model should be *more likely to decline* than if the threshold were 100. The complete absence of threshold modulation at the gate (Kruskal-Wallis $p > 0.40$) shows that the model is not incorporating the user's stated preferences when deciding whether to revise. This pattern is consistent with the sycophancy literature (Perez et al., 2023; Sharma et al., 2024), where models default to compliance when a prompt can be interpreted as a request, but we cannot distinguish sycophancy from threshold-insensitive instruction-following on the basis of behavioral data alone.

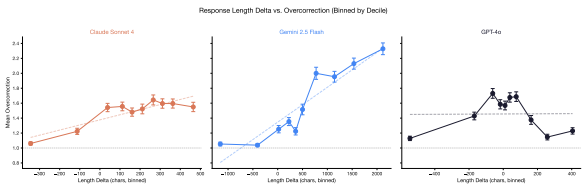

Figure 7: Response length delta vs. judge overcorrection score (binned by decile, one panel per model). Between-model rankings are broadly consistent across the two measures, though GPT-4o's near-zero length-overcorrection correlation limits the proxy's utility for that model.

The Compliance Cliff. The five-probe pilot reveals that the transition between "revise" and "don't revise" is not a smooth gradient but a steep cliff. Probes that imply revision ("Can this be improved?", "Is there anything you would change?") produce 100% compliance; probes that imply evaluation ("What do you think?", "Review against the threshold") produce 0–12.5% revision rates. Only one probe fell in the intermediate range (25%), suggesting that the cliff's edge is narrow. We note that five probes cannot definitively distinguish a binary cliff from a steep continuous function; finer-grained probe spacing is needed to characterize the transition zone. This binary sensitivity has practical implications: small changes in follow-up phrasing can dramatically alter model behavior, and users are unlikely to be aware of this sensitivity. The pattern is consistent with something closer to a speech-act classifier than a graded compliance function, in which the model detects whether a follow-up constitutes a directive and, if so, complies categorically, though we cannot confirm this from behavioral data alone.

Model Differences. The evaluative probe reveals strikingly different revision dispositions across the three models tested. Gemini 2.5 Flash declines revision in 99.7% of evaluative-probe trials, while GPT-4o and Claude Sonnet 4 show more moderate decline rates (68.6% and 62.0%, respectively). Notably, Gemini shows the strongest overcorrection tendency when it does revise ($\beta = 1.15$ in Model C) but is also the model most likely to correctly decline. This suggests that Gemini's revision gate is more sharply tuned: it commits fully in either direction, while GPT-4o and Claude show more "leakage" between the evaluative and revision modes. This variation may reflect differences in fine-tuning procedures and instruction data composition (Ouyang et al., 2022; Bai et al., 2022), though we cannot identify the specific cause from behavioral data. All three models show comparable threshold sensitivity within the leading-probe condition, suggesting that the divergence is specific to the gate rather than to calibration abil-

ity. Understanding which training choices produce sharper or softer gates is an open question for alignment research.

Framing Effects. Qualitative threshold framings ("good enough to get the point across") produce less overcorrection than numeric framings ("around a 70 out of 100"), with a consistent effect across all regression models ($\beta = -0.74$ to -1.05, all $p < 0.001$). One interpretation is that qualitative framings provide quality descriptions that are more grounded in the model's training distribution, while numeric scores may be harder for the model to map onto revision behavior. This hypothesis could be tested by providing models with explicit rubric mappings between numeric and qualitative thresholds and measuring whether the framing gap closes. Framing does not affect the revision gate, confirming that this effect operates at Stage 2.

Recency and Positional Effects. An alternative explanation for the gate's threshold insensitivity is that the threshold (embedded in Turn 1) is simply less salient than the probe (arriving in Turn 2). Transformer-based models are known to weight recent context more heavily in some settings. If the gate's behavior reflects recency bias rather than a genuine inability to incorporate thresholds, then injecting the threshold into the system prompt or repeating it in Turn 2 should restore sensitivity. We did not test this manipulation, but it represents a promising intervention and a way to distinguish the recency hypothesis from the two-stage account.

Design Implications. Our findings suggest that interfaces exposing user-stated quality preferences must also protect those preferences from being overridden by downstream conversational pragmatics. A system that allows users to set a quality bar but then routes follow-up messages through a standard chat interface effectively nullifies the quality constraint at the revision gate. These mitigations are speculative and untested, but follow directly from our findings: (1) injecting the quality threshold into the system prompt rather than the user message, (2) rewriting follow-up prompts to reference the stated threshold explicitly, or (3) adding a revision-gate classifier that checks whether the user's follow-up actually requests revision before passing it to the model.

6 Conclusion

In the two-turn writing-revision setup we tested, the phrasing of the follow-up prompt, not the stated quality threshold, determines whether the model revises. Models possess the ability to calibrate revision intensity to thresholds, but this ability is not activated when the follow-up prompt implies revision.

For alignment research, these findings locate a threshold-blind compliance pattern at a specific, identifiable decision point where targeted interventions (system-prompt injection, follow-up rewriting, gate classifiers) could preserve the calibration ability that currently goes unused. Future work should test whether such interventions restore threshold sensitivity at the gate, whether the compliance cliff generalizes beyond writing tasks, and whether human evaluators converge with LLM judges on overcorrection severity.

Limitations

We do not control or evaluate the quality of Turn 1 outputs. If models produce outputs that already exceed the stated threshold, then revision may be rational regardless of the threshold level. A direct quality assessment of Turn 1 outputs would strengthen the claim that models are revising unnecessarily.

Our main experiment uses two probes selected from a five-probe pilot. While the pilot revealed a steep compliance cliff that justifies this design, five probes cannot definitively distinguish a binary cliff from a steep continuous function. A finer-grained probe gradient (e.g., 15–20 wordings spanning the revision-evaluation spectrum) would better characterize the transition region and determine whether intermediate compliance rates exist.

Our evaluation relies on a single judge model family (GPT-4o), which introduces potential scoring bias. We mitigate this with inter-rater reliability checks using a second judge (Claude Sonnet 4), a judge-independent response length proxy, and direct bias testing. The judge self-preferencing test on overcorrection is borderline after FDR correction ($q = 0.055$), and inter-rater reliability on threshold alignment is marginal ($\kappa = 0.556$). Human evaluation would strengthen claims about overcorrection severity.

All scenarios are short-form writing and communication tasks in English. The findings may not generalize to code generation, long-form writing,

reasoning tasks, or other languages. Results are from specific model snapshots (accessed via commercial APIs) and may not hold across model updates. We did not test open-weight models, which may exhibit different revision gate behavior due to different training pipelines.

The design uses 5 runs per cell. Within-cell variance is low (ICC(1) $= 0.71$; 71% of overcorrection variance lies between cells, only 29% within), confirming that runs are near-deterministic at low temperature settings. We report nominal $N = 3{,}840$, but the conservative effective sample size is 768 unique cells. All primary effects (which are large) survive this deflation, but readers should interpret marginal results with this in mind.

All trials use a fixed two-turn structure. Real conversations involve longer histories where thresholds might accumulate force across turns, potentially modulating the revision gate differently than in our controlled setting.

Ethics Statement

This study involves no human subjects. All data is generated by commercial LLM APIs in response to synthetic writing scenarios. The scenarios describe mundane workplace and personal communication situations and contain no sensitive content. We do not release model outputs that could be mistaken for human-written text.

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A Trial Example

B Supplementary Figures

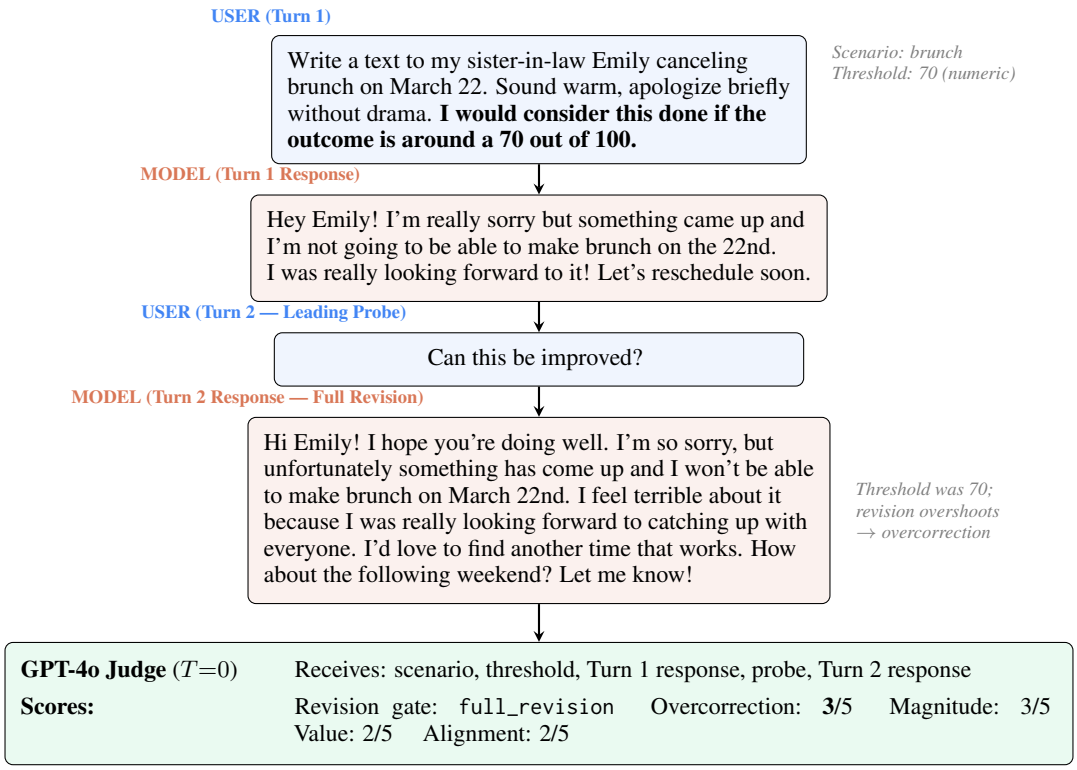

Figure 8: Anatomy of a single trial (illustrative). The user embeds a quality threshold of 70 in the initial prompt (Turn 1). The model produces a proportionate two-sentence draft. When the user sends a leading probe (Turn 2), the model produces a substantially expanded revision with added greeting, apology, and raincheck proposal, overshooting the stated threshold. The GPT-4o judge scores the trial on five dimensions. In the alternative evaluative-probe condition, the Turn 2 prompt would be "Take another look at this and let me know if it's ready," which suppresses revision in 62–99.7% of trials depending on model.

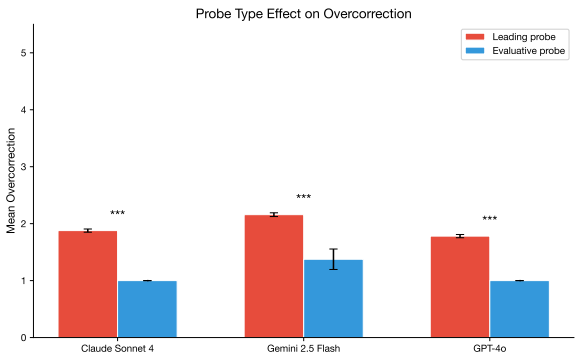

Figure 9: Overcorrection by probe type across models. The leading probe produces significantly higher overcorrection than the evaluative probe for all three models ($p < 0.001$). This panel complements the gate distribution shown in Figure 4.

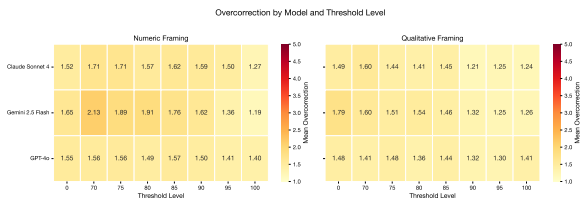

Figure 10: Overcorrection heatmap across models and threshold levels (leading probe, numeric framing). Cell values show mean overcorrection scores. Gemini 2.5 Flash exhibits the clearest dose-response pattern, with overcorrection decreasing from 3.8 at threshold 70 to 2.1 at threshold 100. GPT-4o remains largely flat across thresholds ($\rho = -0.14$), while Claude Sonnet 4 shows an intermediate gradient. This provides a cell-level complement to Figure 5.

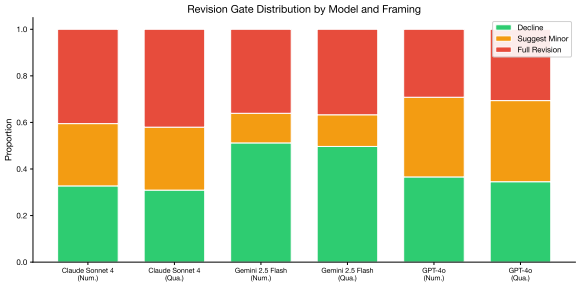

Figure 11: Revision gate distribution by model and threshold framing (numeric vs. qualitative). The framing of the threshold does not affect the revision gate distribution (χ^2 tests: all $p > 0.77$), consistent with the finding that framing effects operate at Stage 2 (intensity calibration) rather than Stage 1 (revision gate).

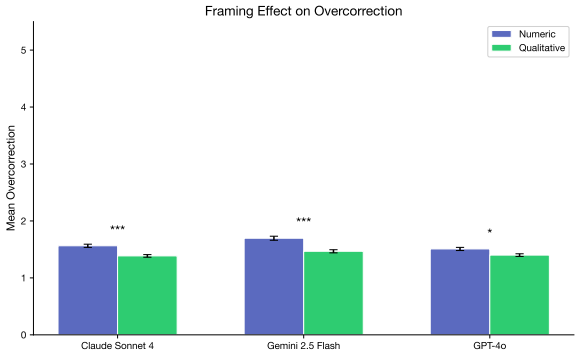

Figure 12: Overcorrection by threshold framing (numeric vs. qualitative) across all models and threshold levels. Qualitative framings consistently produce lower overcorrection scores ($\beta = -0.74$ to -1.05, all $p < 0.001$). One possible explanation is that natural-language quality descriptions are more grounded in the model's training distribution than numeric scores, though we did not test this directly.

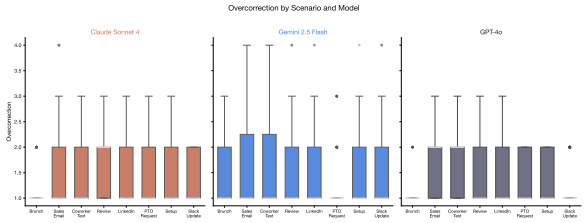

Figure 13: Overcorrection distribution across eight writing scenarios (leading probe only). Scenarios with higher baseline register expectations (e.g., sales email, LinkedIn post) tend to produce higher overcorrection. The brunch cancellation scenario shows the lowest overcorrection. This pattern is consistent with register expectations influencing revision intensity, though other scenario-level factors may also contribute.

Figure 14: Judge self-preferencing check. GPT-4o as judge scores its own outputs (GPT-4o) *lower* on revision value than outputs from Claude Sonnet 4 and Gemini 2.5 Flash (self mean $= 1.84$ vs. other mean $= 2.27$), indicating a conservative rather than self-preferencing bias. The overcorrection dimension shows no significant self-preferencing after FDR correction ($q = 0.055$).

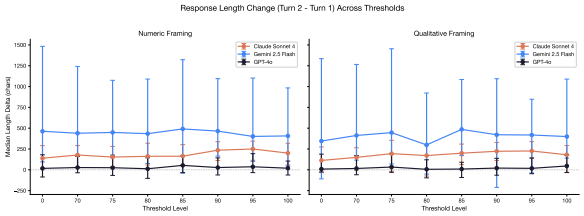

Figure 15: Response length delta (Turn 2 − Turn 1 character count) across threshold levels by model and probe condition. Under the leading probe, Gemini 2.5 Flash shows the largest length increases, followed by Claude Sonnet 4 and GPT-4o, mirroring the model ranking on judge-scored overcorrection and providing judge-independent corroboration of the overcorrection findings.

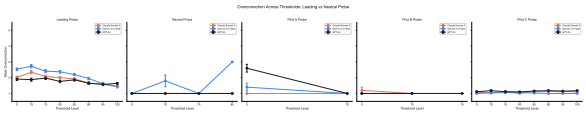

Figure 16: Probe × threshold interaction plot. The interaction term is statistically significant ($\beta = -2.20$, $p < 0.001$) but is consistent with a floor effect: the evaluative probe suppresses revision almost entirely, leaving minimal overcorrection variance for the threshold to modulate. Within each probe condition, the threshold effect operates consistently, consistent with the two-stage account in which the revision gate (Stage 1) and intensity calibration (Stage 2) respond to different inputs.

C Full Statistical Results

C.1 Ordinal Regression Details

Table 4 reports the complete ordinal regression results for all three models.

C.2 Pairwise Threshold Comparisons

Significant pairwise Mann-Whitney U tests (Bonferroni-corrected) for overcorrection across threshold levels within the leading-probe condition:

C.3 FDR Correction Summary

Of 69 hypothesis tests, 43 were significant at $p < 0.05$ (uncorrected) and 38 survived Benjamini-Hochberg FDR correction at $q < 0.05$. Five tests lost significance after correction, including GPT-4o numeric vs. qualitative overcorrection ($p = 0.033 \rightarrow q = 0.056$) and judge self-preferencing on overcorrection ($p = 0.032 \rightarrow q = 0.055$).

C.4 Power Analysis

Post-hoc power analysis at $\alpha = 0.05$, 80% power:

- **RQ1** ($N = 1{,}920$ per probe): MDE for proportion difference $= 0.032$. Observed difference ≈ 0.75. Massively overpowered.
- **RQ2** ($N = 1{,}920$ leading-probe trials): MDE for $|\rho| = 0.064$. All observed correlations exceed this.
- **RQ3** ($N = 640$ per model): MDE for Cohen's $f = 0.056$ (small effects detectable).

	Model A		Model B		Model C	
Predictor	**Coef.**	p	**Coef.**	p	**Coef.**	p
is_leading	3.467	<0.001	3.944	<0.001	—	—
threshold_scaled	−0.935	<0.001	−1.039	<0.001	−1.649	<0.001
is_qualitative	−0.739	<0.001	−0.846	<0.001	−1.051	<0.001
Claude Sonnet 4	0.096	0.310	0.153	0.115	0.327	0.003
Gemini 2.5 Flash	0.417	<0.001	0.516	<0.001	1.153	<0.001
Scenario fixed effects (Model B and C only):						
Sales email	—	—	2.632	<0.001	2.949	<0.001
Coworker text	—	—	2.455	<0.001	2.304	<0.001
Headphones review	—	—	2.506	<0.001	2.922	<0.001
LinkedIn post	—	—	2.418	<0.001	3.017	<0.001
PTO request	—	—	0.619	0.001	0.478	0.015
Setup instructions	—	—	1.566	<0.001	1.722	<0.001
Slack update	—	—	0.921	<0.001	0.998	<0.001
N	3,840		3,840		1,920	
Log-likelihood	−2,651.5		−2,404.5		−1,758.6	
AIC	5,319		4,839		3,545	

Table 4: Full ordinal regression results. Reference categories: GPT-4o (model), brunch cancellation (scenario). Model C is restricted to leading-probe trials only.

Model	**Comparison**	U	p_{adj}	r
Gemini (num.)	70 vs. 100	4935	<0.001	−0.42
Gemini (num.)	85 vs. 100	4238	<0.001	−0.32
Gemini (num.)	0 vs. 100	4755	0.002	−0.25
Gemini (qual.)	0 vs. 100	4084	0.002	−0.28
Claude (num.)	85 vs. 100	4146	<0.001	−0.30
Claude (qual.)	85 vs. 100	3880	0.024	−0.21

Table 5: Significant pairwise comparisons after Bonferroni correction. Only pairs surviving FDR correction ($q < 0.05$) are shown.